

A plan for a World Trade Organization sectoral agreement on semiconductors

Petros C. Mavroidis and Mario R. Palacios

To promote their economic security, or simply to inflict damage on other nations, countries are increasingly weaponising trade policy. To do so, trading nations must sidestep the rules-based system established in the aftermath of the Second World War because this system, in letter and spirit, opposes such behaviour. However, neglecting the rules-based system leads to uncertainty, and uncertainty incentivises trading nations to intensify their pursuit of national economic security, thus undermining the multilateral edifice even more.

It does not have to be this way, however. Based on an analysis of trade in semiconductors (a product market in which there is clear weaponisation of trade policy), we argue that, while a comprehensive negotiation on economic security may be currently unrealistic, a sectoral agreement within the World Trade Organization would be both feasible and appropriate, and could provide a template for regulating economic security. Such an agreement should include non-discriminatory trade in high-end semiconductors and a negotiated list of segments of the covered product market, in which signatories will be allowed to erect barriers to protect economic/national security without being obliged to compensate affected parties.

JEL Classification: K40

Key Words: WTO; semiconductors; economic security

Mario R. Palacios is Managing Director, Government Affairs & Head of International Trade Policy at Applied Materials

Petros C. Mavroidis (petros.mavroidis@bruegel.org) is a Non-resident Fellow at Bruegel

We are indebted to Simon Evenett for sharing his data with us, and to Kyle Bagwell, Emanuel Ornelas, Bernard M. Hoekman, Henrik Horn, Merit E. Janow, Matteo Maggiori, Inu Manak, Chris Miller, Stratos Pahis, Lucio Pench, Niclas Poitiers, Tom J. Prusa, Stephen Redding, André Sapir, Alan Sykes, Reinhilde Veugelers and Jeromin Zettelmeyer for numerous discussions on the issues discussed in this paper. We especially thank Ignacio Garcia Bercero for excellent detailed comments that helped improve our original draft.

Recommended citation:

Mavroidis, P. C. and M. R. Palacios (2026) 'A plan for a World Trade Organization sectoral agreement on semiconductors,' *Working Paper 21/2026*, Bruegel, available at <https://doi.org/10.64153/LXWL4325>

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1 Introduction

The past decade has seen a resurgence of national industrial policy initiatives in the semiconductors market. The rationale for these interventions is not simply rent-seeking in a promising market: it is the proclaimed protection of national and economic security. The United States, European Union, China, Japan, Korea, Taiwan and many other economies have seen increasing government intervention¹. Various instruments are used: subsidies to boost domestic competitiveness or export restrictions to deprive others of critical inputs – or final products, for fear of reverse engineering.

This paper argues that the World Trade Organization is ill-equipped to deal with such challenges as, unless the issue of economic security is brought under the aegis of the multilateral system and acknowledged as a legitimate objective that permits deviations from obligations, trading nations will be forced to continue exercising the ‘outside’ option to achieve their economic security priorities. The ‘outside option’ is the tendency for countries to pursue policies outside of or incompatible with WTO rules, weakening the institution.

The WTO’s relevance has been waning for some time. Its judicial function has been seriously undermined by the Appellate Body being in abeyance, while its legislative arm has only added three agreements since 1995². Now, with the steep increase in measures taken by its members, which are impossible to reconcile with the existing WTO framework, the institution’s policy relevance is waning even faster.

To help redress the current situation, this paper argues that, inspired by past practice, WTO members should opt for a sectoral agreement on semiconductors. This agreement would require WTO members to trade semiconductors in line with multilateral rules, thereby enhancing economic security by promoting non-discriminatory trade without weakening the WTO.

Section 2 provides an overview of the state of world trade today, starting with the Hirschman (1945) argument that trade openness is welfare-enhancing but creates vulnerabilities as well, an oft-forgotten consequence of trade liberalisation. The section also explains why the General Agreement on Tariffs and Trade (GATT)/WTO did not immunise itself against this risk.

Section 3 focuses on the semiconductors market and explains why it serves as the most appropriate case for nations to achieve economic security. Section 4 assesses the way nations have segmented the semiconductor market in pursuit of economic security and highlights why international law as it currently stands cannot guarantee legitimate concerns for economic security. Section 5 lays out a proposal for a sectoral agreement on semiconductors, which is both feasible and has the main merit of preserving and promoting non-discriminatory trade liberalisation. Section 6 recaps the main conclusions.

This paper contributes to an expanding literature on industrial policy, geoeconomics

With the steep increase in measures that are impossible to reconcile with the WTO framework, the institution’s policy relevance is waning fast

1 OECD data leaves no room for doubt in this respect. See OECD, ‘MAGIC Database of Industrial Subsidies’, <https://www.oecd.org/en/data/dashboards/magic-database-industrial-subsidies.html>.

2 The ITA (Information Technology Agreement), the Fisheries Subsidies Agreement and the Agreement on Trade Facilitation (TFA).

and trade integration. Clayton *et al* (2024) provided a framing paper of the issues involved. We distinguish between two strands in the literature: papers such as Rotunno and Ruta (2025), who discussed the effects of the responses to trade-segmenting measures adopted first by the United States; and papers such as Carvalho *et al* (2025) and Mattoo *et al* (2024), which discuss the effects of unilateralism (such as the 2025 tariffs adopted by the Trump Administration) on the multilateral trading system. This paper belongs to the latter category.

2 A history of economic security in theory and practice

In 1748, the French political philosopher Baron de Montesquieu argued in *De l'esprit des lois* (*The Spirit of Laws*) that trade civilises communities:

*On peut dire que les lois du commerce perfectionnent les mœurs, par la même raison que ces mêmes lois perdent les mœurs. . . . L'effet naturel du commerce est de porter la paix. Deux nations qui négocient ensemble se rendent réciproquement dépendantes . . . toutes les unions sont fondées sur des besoins mutuels*³.

With these words, Montesquieu expressed the idea of '*doux commerce*', or mutual interest and contributing to peace as the lynchpins of the civilising force of trade. His ideas found support in academia and among policymakers. Cordell Hull, US Secretary of State during the Franklin D. Roosevelt administration (mid-1930s to mid-1940s), espoused '*doux commerce*'. Hull's deep conviction that trade dovetails with peace was the driving force behind the advent of the GATT⁴.

The first to cast doubt on the civilising influence of trade integration was Hirschman (1945). He claimed that trade openness is welfare-enhancing but comes at the cost of increasing vulnerability. His focus was on the attitude of mercantilists – specifically of Germany in the pre-Second World War era. Hirschman identified two distinct routes through which international trade can affect a state's power: the so-called 'supply effect' and the 'influence effect'. The former refers to the effect of imports (goods and services) on a country's overall trade position; the latter, to the ability of a state to interrupt its own exports and imports. Building on these two effects, Hirschman claimed that there is a link between gains from trade and dependence on trade, meaning that as the benefits from trade increase, so too does dependence on it. His critique was a precursor to the later

3 De l'esprit des Lois, IV, XX, 1, 2. Translation: "We can say that the laws of commerce perfect morals, for the same reason that these same laws destroy morals.... The natural effect of trade is to bring about peace. Two nations which negotiate together make each other mutually dependent ... all unions are based on mutual need". A translation of 'Republican Government' from Montesquieu (1748) is available at <https://press-pubs.uchicago.edu/founders/documents/v1ch4s2.html>.

4 Hull (1948) included extensive references to his battles with the other members of the administration.

concept of the weaponisation of international trade by those with sufficient bargaining power. This link to power made the trade-integration process vulnerable, in Hirschman's view. In a similar vein, Viner (1948) demonstrated that the ability to separate the gains from trade from the exercise of power was not a unanimously accepted proposition.

Today, much is written about the weaponisation of trade policy, the emergence of trade chokepoints and the ensuing risks. These papers analyse how economic interdependence is routinely leveraged by states (Clayton *et al*, 2024; Farrell and Newman, 2019; Fishman, 2025). Conceptually, however, the weaponisation debate has not evolved. For example, McLaren (1997) wrote long before the advent of geoeconomics about 'Judge Bowker's objection'. Inspired by a Canadian story, he showed how trade liberalisation between two countries (a large country and a small one, ie the US and Canada), assuming irreversible investments, could lead to the smaller country ending up worse off, than if it had opted for autarchy. The reason put forth is that investors in a liberal trade environment will invest in the export sector and thus make the smaller country dependent on trade with the larger partner. McLaren (1997) argued that, in this scenario, the smaller country's bargaining power is undermined⁵. Indeed, McLaren's prediction came to pass a few decades later, when the US under President Donald Trump began weaponising trade against its smaller neighbour.

The preconditions for the weaponisation of trade to succeed are concentrated production and low substitutability. The aggressor should not be reliant on supplies from the target country: it must have alternative, stable supply chains, whereas the target country must be heavily dependent on the aggressor for key imports. The US offers an obvious example with the pressure on Dutch firm ASML (Advanced Semiconductor Materials Lithography) to stop selling machinery to China⁶. ASML cooperated but the US administration was willing to escalate its threats of retaliation in the case of non-cooperation.

2.1 The General Agreement on Tariffs and Trade (GATT)

GATT was Chapter IV of a wider negotiation that would lead to the establishment of the International Trade Organization (ITO). Chapter V of the ITO, entitled 'Restrictive Business Practices'⁷, included nine provisions, some addressing head on the concentration of market power. Article 46.1 of the ITO Charter stated:

Each Member shall take appropriate measures and shall co-operate with the Organization to prevent, on the part of private or public commercial enterprises, business practices affecting international trade which restrain competition, limit access to markets, or foster monopolistic control, whenever such practices have

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⁵ McLaren's analysis was prompted by a pamphlet by a Canadian judge (Judge Bowker) who had anticipated that by signing the US-Canada FTA (free-trade agreement), Canada would be worse off than otherwise because Canadians would be making irreversible investments to profit from trade integration with the US.

⁶ Toby Sterling, 'ASML CEO says US restricts servicing some China equipment, won't hurt earnings', *Reuters*, 24 April 2024, <https://www.reuters.com/technology/asml-ceo-says-us-restricts-servicing-some-china-equipment-wont-hurt-earnings-2024-04-24/>.

⁷ Thus, while GATT (Chapter IV) focused on state practices, Chapter V was about disciplining private practices that segmented trade.

harmful effects on the expansion of production or trade and interfere with the achievement of any of the other objectives set forth in Article 1⁸.

Article 46.2(d) went beyond the GATT and sanctioned (besides export restrictions; these are banned in both agreements) *"limiting production or fixing production quotas"*. Finally, Article 46.3(g) included an anti-circumvention provision, stating clearly that the ITO would have jurisdiction to assess *"any similar practices which the Organization may declare, by a majority of two thirds of the Members present and voting, to be restrictive business practices"*.

Members were obliged by Article 50 to avoid the restrictive business practices mentioned in Article 46 (including limiting production, price fixing, exclusion of certain enterprises, price discrimination, etc). Article 49 recognised the right of the ITO to conduct studies into these restrictive business practices and recommend actions to bring them into compliance with members' obligations.

With the ITO ultimately failing (as a result of the refusal of the US administration to submit it to Congress for approval in the early 1950s, fearing that it would transfer too much sovereignty to the international level), the GATT was left without a strong framework. Article XXIX of the GATT linked to the ITO, but it is admittedly a weak obligation. In the absence of antitrust law disciplines, the fight against concerted practices became a question for national law. Not all jurisdictions adopted antitrust laws at the time, though. GATT members were not, however, totally helpless. The GATT framework was more lenient towards non-cooperative behaviour and concession-extraction: subsidies were largely unconstrained and there were few if any compromises on existing export duties⁹. Nevertheless, the original hegemon (US), and subsequent dominant players (US, EU) made moderate use of this flexibility¹⁰.

This begs the question of why past behaviour, when not legally constrained, still leaned towards cooperation. Two basic reasons explain why.

First, security concerns about the Soviet Union and its satellites reduced GATT members' motivation to weaken the alliance. Both Jackson (1969) and Irwin *et al* (2008) argued that the GATT was a contractual arrangement between the like-minded winners of the Second World War who had become both military and economic allies. The original GATT members either belonged to the North Atlantic Treaty Organization (NATO), or were

8 *"Increase the production, consumption and exchange of goods"* was one of the objectives explicitly mentioned in the body of Article 1 of the ITO Charter.

9 There were few concessions anyway, and most of them were offered by small players who could not have affected terms of trade through similar measures, see WTO, 'Article II, Schedules of Concessions', https://www.wto.org/english/res_e/publications_e/ai17_e/gatt1994_art2_gatt47.pdf.

10 In practice none of them imposed export duties. Over the years, the US adopted a hostile attitude towards subsidies. The EU disallowed member states' subsidies in principle (in order to avoid distortions of its internal market), whereas most EU-level subsidies were concentrated in the farm markets.

friendly to NATO forces¹¹. Restricting the original GATT to such like-minded countries was a clear priority for negotiators¹². Relations between these countries and the Soviet Union and its satellites was regulated through the Coordinated Committee for Multilateral Export Controls (COCOM) initially, and the Wassenaar Agreement later on. The major economies thus behaved like Kindleberger's (1973) hegemon¹³, rather than as Hirschman (1945) had feared.

The second reason is that the level of economic integration was left to the sovereign choice of the members¹⁴. A safeguard clause was included to allow for breathing space in case the original calculations proved to be wrong¹⁵. Finally, the GATT included a national security clause (Article XXI). This however was almost never used, as the GATT included allies, originally at least. The 'Eastern bloc' countries that joined the GATT subsequently (Hungary, Poland, Romania, Yugoslavia) were too small to threaten the incumbents, and anyway, because of the nature of the commitments they assumed, were living outside the GATT disciplines anyway. COCOM continued to be relevant when it came to trade between them and the GATT incumbents.

The GATT regime thus provided an adequate safe haven for concerns relating to the national security, owing more to the attitude of the participants than to institutional protections against the weaponisation of trade policy.

2.2 The end of history

The WTO was negotiated during the 'apex' of liberalism between 1986-1994. Halfway through the negotiation, the Berlin wall fell – the most symbolic event signalling the end of the Cold War. Francis Fukuyama (1992) called it the "*end of history*" and the advent of a unipolar world. Previous enemies and rivals became increasingly like-minded, all rushing into the multilateral house. Consequently, the Uruguay round negotiators (the Uruguay round started in 1986) did not amend the GATT Article XXI to include a discussion on economic (in addition to national) security).

Reich's discussion of the perils of techno-nationalism¹⁶ did not initially gain much

11 This statement might seem an oddity in light of the inclusion of Brazil, China and Czechoslovakia among the negotiators of the original GATT. Brazil's then president, Eurico Gaspar Dutra, outlawed the Communist Party and signed the 1947 Rio Treaty, a security agreement with the US, while opening on US investment. US-Czechoslovakia relations were in a delicate state, and the latter had even accepted an initial offer to join the Marshall Plan, which it later refused following pressure by Stalin. A coup in Czechoslovakia took place in March 1948, when the GATT had already been negotiated. The US was on good terms with the Nationalist Kuomintang, but China was going through a civil war at that time, and left practically no footprint in the GATT negotiations.

12 The Soviet Union satellites that joined the GATT (Hungary, Poland, Romania, Yugoslavia) were too small to be a military concern for the big GATT players. The same is true also for Czechoslovakia, which was an original GATT member and changed sides three months after the entry into force of the GATT.

13 Kindleberger stated that it is in the interest of a hegemon (which is not facing a competitive fringe) to assume the requisite cost and invest in a multilateral regime that smaller nations will find it in their interest to adhere to.

14 Clayton *et al* (2024) advanced an alternative and yet similar explanation, when arguing that Hirschman's work precedes the advent of the international liberal order, which made weaponisation of trade policy a remote possibility.

15 Bagwell and Staiger (2002) provided an authoritative account on the economics of trade agreements and the GATT.

16 Robert Reich, 'The Rise of Techno-Nationalism', *The Atlantic Monthly*, May 1987, <https://www.theatlantic.com/magazine/archive/1987/05/the-rise-of-techno-nationalism/665772/>.

prominence. Reich had warned that technology was “leaky” and that policies aiming to contain technological developments within national frontiers were bound to fail. Shortly thereafter however, it all changed. China’s accession to the WTO in 2001 proved especially tough to manage. Ideological heterogeneity within the membership increased substantially, but this was momentarily overlooked after the fall of the USSR. China was accused of not becoming a market economy¹⁷, and China was disillusioned by how the West (and especially the US) handled of the 2008 global financial crisis. The COVID-19 pandemic further eroded trust, as WTO members turned inwards to ensure their own financial security and adequate supplies of personal protective material. By this time, it was too late to revisit the WTO’s legal arsenal. The consensus-based approach to adopting decisions proved to be a major impediment to action.

Though the COVID-19 pandemic passed, the issues raised by China’s rapid growth did not. The first Trump administration imposed unilateral tariffs on imports from China before negotiating a WTO-inconsistent agreement. This, the so-called Phase 1 Agreement, was never implemented by China, leading to the re-imposition of tariffs. The Biden administration and second Trump administration adopted roughly the same attitude towards China, imposing export controls on semiconductors and more generally on high-tech products. Mavroidis (2025) discussed how the Biden administration made extensive use of the ‘direct product’ rule, which subjects foreign-produced items to US jurisdiction if they are a direct product of specific, US-origin technology, software or are manufactured by a facility that is itself a direct product of such technology.

Additionally, the US initiated a series of Section 232 investigations that led to the imposition of safeguards on steel products from some WTO members¹⁸. A series of WTO panels outlawed these measures, yet the US refused to change course. Uncertainty about future US trade policy continued to increase, especially as the Biden administration did not remove any of the tariffs on China.

2.3 The present moment

The environment that was supposed to provide certainty for international trading became increasingly wobbly (Handley and Limão, 2017). The US-China rivalry is a key reason, but not the only reason for this.

Herz (1950) claimed that one state’s actions to increase its security – which could take different forms, from building up military strength to forming alliances to investing in strategic sectors – could be perceived as a threat by other states, which would then in turn adopt similar strategies to increase their own security. This theory applies to economic security as well. Nations have subscribed to Herz’s ‘security dilemma’ (Figures 1 and 2).

17 Mavroidis and Sapir (2021) explained why accession to the WTO by itself could not act as catalyst for a change of regime in China.

18 Section 232 of the US Trade Expansion Act of 1962, as amended (19 U.S.C. §1862) allows the president to impose restrictions on goods imports or enter into negotiations with trading partners if the Secretary of Commerce determines, following an investigation, that the quantity or other circumstance of those imports “threaten to impair” US national security.

Figure 1: Protecting and liberalising measures at the global scale

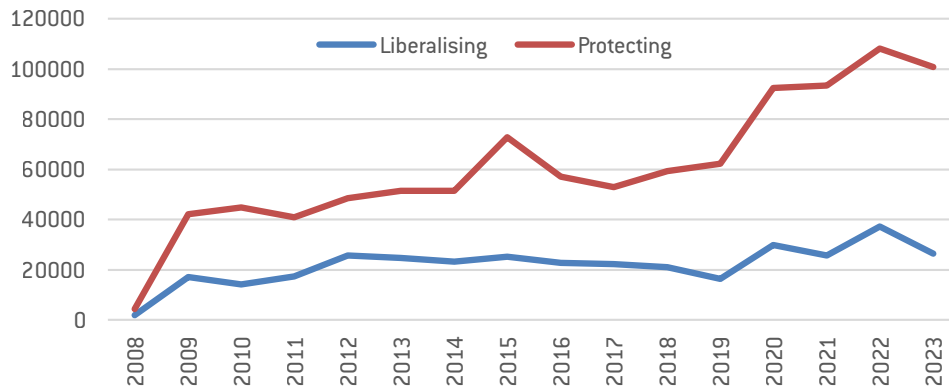
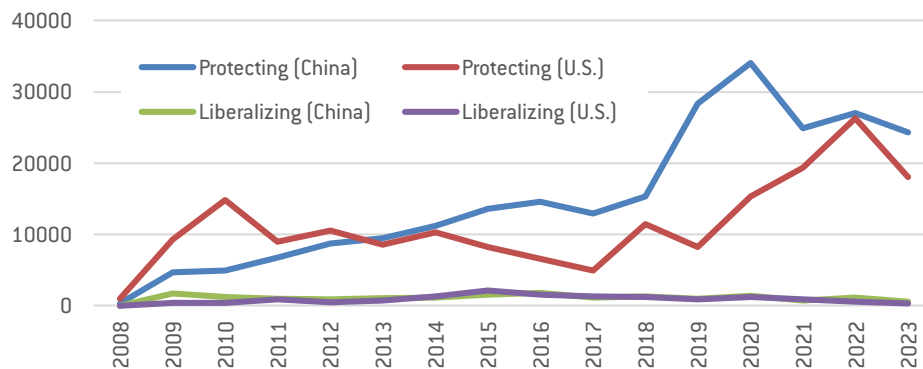


Figure 2: US and Chinese protectionist and liberalising measures

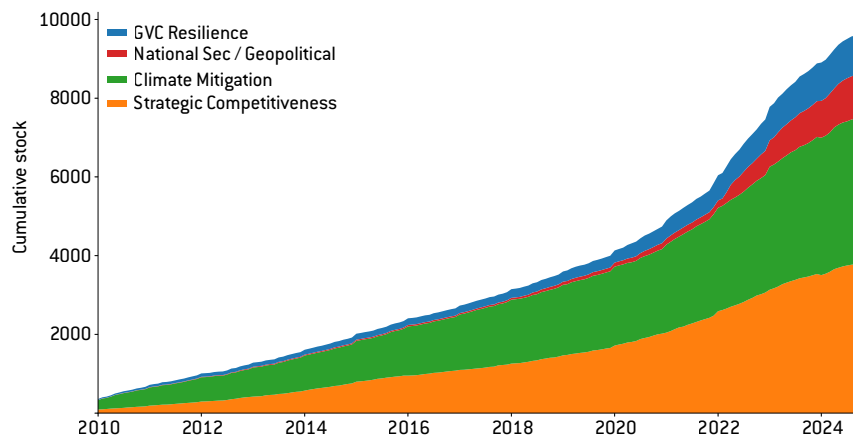


Source: Evenett *et al* (2024).

Evenett *et al* (2024) offered their own coding of trade-liberalising as opposed to trade-obstructing measures. What mattered for them is the potential effect of adopted measures.

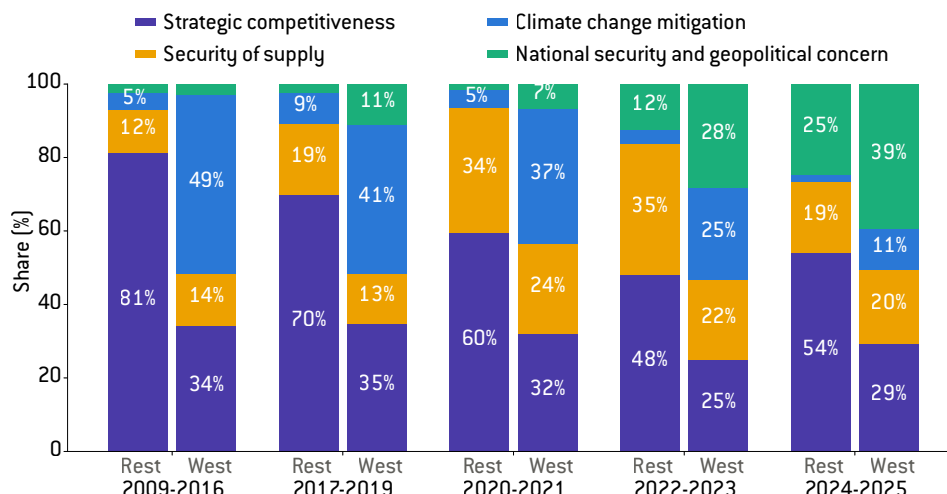
The world's two largest economies were not alone in their drift towards protectionism, and soon other countries began to mimic their policies. Evenett *et al* (2024) showed the rationale for protectionist measures enacted by countries since 2010.

Figure 3: Revealed objectives of trade-restricting measures



Source: Evenett *et al* (2024). Note: cumulative stock of measures. For measures with multiple motives, each motive is given equal weight.

Figure 4: Stated motives for industrial actions by period



Source: NIPO by Global Trade Alert, December 2025. Note: 'The West' refers to Australia, the EU, the G7 and South Korea.

The trend continues to strengthen. As Jakubik *et al* (2026) stated:

"... 2020 marks a structural shift in the number of industrial policy interventions.... The COVID-19 pandemic, commencing in January 2020, was clearly catalytic, but policy activity still remains elevated well after countries have exited the pandemic.... In other words, prior to 2020, countries implemented relatively more industrial policies targeting products in which they already had a strong global market presence, but this has since switched, implying a shift in focus to new and less established sectors. An example would be a country that previously supported firms in heavy industries refocusing its support to downstream, higher value-added manufacturing".

Jakubik *et al* (2026) added:

"Prior to 2020, countries implemented relatively fewer industrial policies targeting products imported from geopolitical rivals, but this has since reversed, suggesting growing interest in reducing geopolitical risks from dependencies".

The choices of China, the US and EU are telling. China is a clear example of acting alone. Through the Belt and Road Initiative (BRI) it has influenced the policy choices of recipient countries as a *quid pro quo* for Chinese loans for infrastructure and energy systems upgrades. It has brought Chinese investment into resource-rich countries in Africa and elsewhere¹⁹.

The US policy is a mix: it has certainly acted alone under President Trump, especially in April 2025 when it unilaterally decided to do away with its most basic WTO commitments by raising US tariffs to unprecedented levels. Trump's second administration has attempted

¹⁹ Niarchos (2025) provided an excellent account of Chinese penetration into the African market.

to build alliances with specific players in specific strategic markets. Of immediate interest is the '*pax silica*' initiative²⁰, which aims to strengthen and to coordinate global value chains (GVCs) for advanced technologies, including semiconductors, artificial intelligence (AI), infrastructure, critical minerals, advanced manufacturing and logistics. Australia, Greece, Israel, Japan, Qatar, South Korea, Singapore, the United Arab Emirates and the United Kingdom have signed it. India joined on 20 February 2026²¹.

The trade truce that China and the US signed on 12 May 2025 was not a binding international agreement. Using '*pax silica*', the US has led efforts to reduce dependency on China's near-monopoly on critical minerals and rare earth elements, which it has been weaponising. The US has initiated the Forum on Resource Geostrategic Engagement, with the aim of creating a preferential trade zone with enforced price floors for critical minerals. It is the successor to the 2022 Minerals Security Partnership²², another US-led international initiative that aimed to secure critical mineral supply chains and reduce dependence on China²³. Fifty-four countries participated in the 2026 Critical Minerals Ministerial to discuss a plan of action. President Trump has also announced the 'Project Vault', a plan for stockpiling REEs, that will be funded with a \$10 billion loan from the US Export-Import Bank, and nearly \$1.67 billion through private capital²⁴.

The EU's policy is also a mix. It has adopted frameworks for state aid to boost strategic sectors²⁵ and participates in the Trump administration's initiatives discussed above. The EU's target, formalised in the Critical Raw Materials Act (Regulation (EU) 2024/1252), is for at least 25 percent of its annual consumption of strategic raw materials, including rare earth elements, to be sourced from domestic recycling by 2030. Earlier industry initiatives, such as those from the European Raw Materials Alliance, aimed for 20 percent of European rare earth needs for magnet manufacturing to be met via recycling by 2030, a goal supported by a €1.7 billion investment plan to boost urban mining. And the EU has concluded free-trade agreements with Australia, India and MERCOSUR²⁶ to find reliable trade partners. The EU, though often plagued by internal divisions, also has enough economic weight to weaponise trade, should it choose to do so (Barth and Eisl, 2026).

Regardless of whether they act in concert or alone, it is quite clear that reducing uncertainty through diversification is a key objective for the EU and US, whereas China addresses the risk of uncertainty mainly by trying to secure supplies through investment

20 See U.S. Department of State, 'Pax Silica', undated, <https://www.state.gov/pax-silica>.

21 See Government of India press release of 20 February 2026, 'India Joins Pax Silica Initiative, Signs the Joint Statement on India-U.S. AI Opportunity Partnership', <https://www.mea.gov.in/press-releases.htm?dtl/40800/India+Joins+Pax+Silica+Initiative+Signs+the+Joint+Statement+on+IndiaUS+AI+Opportunity+Partnership>.

22 On governance of critical minerals in more general terms, see Sasmal (2026).

23 Among the most important critical minerals are a group of seventeen metals known as rare earth elements (REEs). They are a key input into the production of semiconductors.

24 Crebo-Rediker and Khan (2026) called for alternative strategies that the US should adopt: instead of replicating China, it should adopt innovative strategies, such production of rare earth-free and reduced-content magnets. They also argue that the US should build alliances to embed innovation-led mineral security, as going alone in this context is not an option. See also Michael Froman, 'US Allies Aim to Break China's Dominance', *Expert Take*, 6 February 2026, Council on Foreign Relations, <https://www.cfr.org/articles/u-s-allies-aim-to-break-chinas-critical-minerals-dominance>.

25 The payment of aids is at the state level, the EU limiting itself to providing the rulebook stating the conditions for lawful subsidisation.

26 Argentina, Brazil, Paraguay, Uruguay.

in resource-rich countries. Hirschman (1945) would feel vindicated after all these years. He feared rightly that trade openness creates vulnerabilities.

2.4 Why economic and national security, not economic/national security

Economic and national security are overlapping concepts, but they are not synonymous. An abstraction of the concept is as follows: state A and state B are asymmetrically exposed to international trade. Assume A trades only with B, whereas B trades with the world. A will behave in accordance with '*doux commerce*': it is not in its interest to upend its relationship with B, for B is the exclusive supplier of and market for its goods. A will not risk its national security by engaging in warlike behaviour towards B, even though A is economically very insecure: if B becomes less open, A will suffer a substantial welfare reduction.

On the other hand, B is economically secure because of its diversification of supplies and export markets. B can afford to behave as it pleases and even adopt belligerent behaviour towards another nation. Its dependence on a specific source is minimal. Economic security is best served when trade is conducted in smaller volumes with many suppliers. In other words, whereas national security is about promoting the existence of a state, economic security is about protecting the national GDP.

2.5 The resulting uncertainty

The WTO is a rules-based system which all its members have agreed to. In such a system there is no room for deviant behaviour, even by those with large bargaining power. Kindleberger (1973) explained why it is in the interest of a hegemon to maintain its commitment to the rules, and Carvalho *et al* (2025) provided a theoretical framework supporting this. A rules-based system provides certainty via the predictability of members' behaviour. This is not to say that 'rules-based' systems are complete contracts. But competition will occur in accordance with pre-defined rules. When nations operate outside of the rules, uncertainty creeps in. The US has been the most frequent culprit in this regard, with its practices being found to be inconsistent with its WTO commitments by various panels. The legality of the practices of others has yet to be challenged.

Caldara *et al* (2020) developed the Trade Policy Uncertainty (TPU) index. It tracks the frequency of terms related to 'trade policy' and 'uncertainty' appearing in major newspapers. It is clear from their work that the Uncertainty Index has reached new heights after January 2025, when the second Trump Administration arrived at the White House. To this day, it shows no signs of retreating to its pre-2025 levels.

The 20 February 2026 US Supreme Court decision to outlaw the 2 April 2025 Liberation Day tariffs that President Trump enacted by invoking the International Emergency Economic Powers Act (IEEPA)²⁷ was immediately followed by Trump's decision to invoke Section 122, allowing him to impose a flat, non-discriminatory tariff of 15 percent²⁸. This decision affects the deals that various nations signed with the US, and it remains to be seen at time of writing how this will be reconciled. The risk of further uncertainty in global trade is thus quite substantial.

WTO economists have revised their trade projections lowering their estimate for 2026 to 0.5% (down from the previous forecast of 1.8%). Trade growth is expected to slow in 2026 as the global economy cools and as the full impact of higher tariffs is finally felt for a full year. The WTO's forecast for world commercial services trade in volume terms has also been revised in light of recent developments. Although not directly subject to tariffs, services trade can be affected indirectly through links to goods trade and output. Services export growth is now expected to slow from 6.8% in 2024 to 4.6% in 2025, and down to 4.4% in 2026. "Slowbalization", e.g., the prolonged deceleration of slower trade growth, this more stagnant phase to put it in blunt terms, is the natural consequence of the reigning uncertainty regarding the conduct of trade policies by especially the bigger players.

2.6 Uncertainty, 'slowbalisation' and no hegemon in sight

The weaponisation of trade policy has created uncertainty which has in turn led to 'slowbalisation', or a backing off from globalisation (WTO, 2025). Despite multiple crises, the world economy has shown remarkable resilience. Devesa *et al* (2026) showed that in 2024 and 2025 trade grew faster than the world economy in general, even though trade between geopolitically distant players (eg trade between the US and China) shrank substantially. But there are countervailing forces: artificial-intelligence-related shipments have increased considerably; China has managed to find new markets (since the US market has been closed in part to this exports); the US has increased its volume of imports from other South East Asian countries; and finally, President Trump does trade policy for "*optics, not for outcomes*"²⁹. A number of announced tariff-restricting measures were either not implemented, or simply abandoned.

At time of writing, uncertainty persists. Certainty is provided only by playing in line with the multilateral rulebook; a less and less common practice in international trade.

Linking trade to power (for those who can afford it) has provoked a quest for economic security by players with lower bargaining power. The WTO as-is does cannot adequately deal with this new policy priority, and so should be amended. Is it possible? Legislative amendments require consensus to be enacted, and consensus is in short supply in the

27 Supreme Court of the United States, 'Learning Resources, Inc., et al. v. Trump, President of the United States, et al.', https://www.supremecourt.gov/opinions/25pdf/24-1287_4gcj.pdf.

28 See The White House, 'Fact Sheet: President Donald J. Trump Imposes a Temporary Import Duty to Address Fundamental International Payment Problems', 20 February 2026, <https://www.whitehouse.gov/fact-sheets/2026/02/fact-sheet-president-donald-j-trump-imposes-a-temporary-import-duty-to-address-fundamental-international-payment-problems/>.

29 As put by Richard Baldwin, 'Why Didn't Trump Tariffs Wreck the World Trade System', Richard Baldwin Substack, 28 March 2026, <https://rbaldwin.substack.com/p/why-didnt-trumpian-tariffs-wreck>.

current global environment. Someone needs to take the lead in proposing reforms; the GATT regime was possible because of the efforts of a hegemon. Kindleberger (1973) understood the hegemon to be a super-dominant player that faces no competitive fringe. Hegemonic stability theory posits that a hegemon will opt for a rules-based system because of the potential hold-up problem³⁰. Power might be dispersed among many small players, but they might still be unwilling to participate in a power game from which they have nothing to win.

Carvalho *et al* (2025) attempted to formalise this theory, adding some important nuances. They distinguished between a hegemon, dominant players and subordinates³¹. While the hegemon and small players prefer a rules-based system, the preference of dominant players is for a power-based system. Dominant players face a prisoner's dilemma: while their best option would be to jointly pursue and support a rules-based system, in the absence of any leader their next-best policy is to go it alone (a power-based system).

Coordination costs should not be underestimated. Most-favoured nation (MFN) and compulsory third-party adjudication are probably the two quintessential features of the multilateral system. In proposals by China, the US and the EU on WTO reform³², two of the three tend to agree or they do not agree at all. The EU and US have espoused 'conditional' MFN status as the recipe for future trade integration. China continues to support MFN as it currently stands. China and the EU on the other hand, are supporters of compulsory third-party adjudication within the WTO, which the US opposes. This leaves the EU agreeing sometimes with the US, sometimes with China, and the US and China disagreeing on everything.

The WTO is thus facing a 'Kindleberger trap'³³. The question arises of whether this the end of multilateralism. Should the world trading community abandon the multilateral game and retreat to regional integration³⁴? We believe not. But first, it is time to delve into the specific of the semiconductor industry.

30 This is a fundamental economic and contract theory concept according to which two parties could work together efficiently but refrain from doing so because of the fear that one party will exploit the relationship to gain all the bargaining power and profits after an initial, unrecoverable investment has been made.

31 The hegemon in their model faces no competitive fringe. The dominant players have bargaining power but face no competitive fringe from each other. Subordinates have no bargaining power.

32 WTO Docs. WT/GC/W/989 of 18 February 2026 (China); WT/GC/W/984 of 15 December 2025 (US); and WT/GC/W/986 of 21 January 2026 (EU).

33 Joseph S. Nye, 'The Kindleberger Trap', *Project Syndicate*, 9 January 2017, <https://www.project-syndicate.org/commentary/trump-china-kindleberger-trap-by-joseph-s-nye-2017-01>.

34 Pahis (2025) made a persuasive case for a plurilateral agreement to address the issue of national security in an effort to preserve the relevance of the multilateral process.

3 The semiconductor industry

Semiconductors are a dual-use good with both commercial (civilian) and military uses. Semiconductors are also a key component in the AI industry. Demand for AI infrastructure, high-performance memory chips and cloud computing has led some analysts to state that the semiconductor industry is experiencing a new business super-cycle³⁵.

2026: Semiconductor Market Surpasses USD 1.5 Trillion Driven by Extraordinary Memory Expansion

Following exceptionally strong results in late 2025 and early 2026, the global semiconductor market is now projected to grow 90 percent in 2026, reaching USD 1.51 trillion.

The sharp upward acceleration is overwhelmingly driven by the Memory segment, which is forecast to surge by around 250 percent year over year, reaching more than USD 800 billion in 2026. Continued strong demand for AI infrastructure, high-bandwidth memory (HBM), and accelerated computing platforms remains the primary growth catalyst for the semiconductor industry. Logic is expected to remain another major contributor, growing 37 percent in 2026.

Other product categories are forecast to expand at more moderate rates, reflecting a broader industry growth: Microprocessors 20 percent, Analog 10 percent, Discrete Semiconductors 8 percent, Sensors and Optoelectronics with 3 percent.

Regionally, all major markets are projected to show strong growth. The Americas are expected to more than double in 2026 with 112 percent growth, driven by the concentration of AI-related semiconductor demand and cloud infrastructure investments. Asia Pacific is forecast to grow 87 percent, while Europe and Japan are projected to grow 58 percent and 28 percent respectively.

2027 Outlook: Growth Continues Above Historical Industry Trends

For 2027, WSTS forecasts the global semiconductor market to grow a further 27 percent, reaching approximately USD 1.9 trillion.

Memory is again expected to lead market expansion with projected growth of 32 percent, while Logic is forecast to increase by 27 percent. Continued deployment of AI systems, advanced computing infrastructure, and expanding semiconductor content across end markets are expected to support ongoing industry momentum. All geographic regions are projected to continue growing in 2027, led by the Americas and Asia Pacific.

35 Resilinc, 'AI Has Sparked a Supercycle, but Can Semiconductor Supply Chains Keep Up?' 3 December 2025, <https://resilinc.ai/blog/ai-has-sparked-supercycle-can-semiconductor-supply-chains-keep-up/>.

3.1 Unpacking the semiconductor industry

Semiconductors can be memory chips (ie DRAM, NAND³⁶) which store data and programs on computers and/or computing systems and servers, or logic chips (ie CPUs, GPUs, FPGAs, NPUs³⁷) which process data. There are also ASICs (application-specific integrated circuits), which are single purpose, and SOCs (system-on-chip). Alternatively, analogue chips perform repetitive routines, for example barcode scanning, while SOCs integrate several chips and circuits into a single chip device, usually combining graphics, audio, camera, video and Wi-Fi.

Undergirding the vast multitude of semiconductor device types is a complex network of chip designers, manufacturers and fabricators. The principal business models are fabless chip designers, foundries and integrated design manufacturers (IDMs). Notable examples include Nvidia (fabless), TSMC (foundry) and Micron (IDM). These business models and firms are supported by a global network of design software (EDA), manufacturing/assembly equipment, testing and packaging, raw materials and components suppliers.

Today, the major semiconductor industry market segments are:

- Chip design (firms that are either fabless or integrated device manufacturers who design their own chips) – top suppliers are headquartered in the United States, Taiwan, South Korea, the European Union, and China.
- Foundries (firms that only fabricate chips on behalf of customers on a contract basis, foundries work with “fabless” companies—businesses that design chips but outsource production) – top suppliers are headquartered in Taiwan, South Korea, the United States, and China.
- Integrated Device Manufacturers or IDMs (firms which design and manufacture/fabricate) – top suppliers are headquartered in the United States, Japan, the European Union, and China.
- Memory (firms which only design and manufacture/fabricate memory chip devices, such as DRAM, and NAND) – top suppliers are headquartered in South Korea, the United States, Japan, and China.
- Electronic Design Automation (commonly referred to as “EDA”, are firms which only provide software and hardware used to design, simulate, verify, and manufacture semiconductor chips) – top suppliers are headquartered in the United States, the European Union and Japan.
- Wafer Fabrication Equipment or “WFE” (firms in this category design and manufacture the specialized equipment used in semiconductor device fabrication facilities) – top suppliers are headquartered in the United States, the European Union, and Japan.

3.2 Legacy and advanced semiconductors

When assessing the critical parts of the semiconductor industry, policymakers have

36 DRAM refers to Dynamic Random Access Memory; NAND refers to ‘Not AND’ – a type of Boolean logic gate that is fundamental to digital electronics.

37 Central processing units, graphics processing units, field-programmable gate arrays and neural processing units.

labelled device types as either 'legacy' or 'advanced' (see for example Goujon *et al*, 2024). They ignore the function of each device type. Semiconductor devices can be categorised as 'analogue' or 'digital' integrated circuits. Analogue devices account for about 10 percent of the chip market value. These devices process discrete 0s and 1s for data processing and storage. Logic and memory chips are 'digital' integrated circuits. About one-third of the semiconductor market comprises leading-edge logic chip, one-third is memory devices and the remaining third is legacy logic, analogue and power devices. Commonplace labels such as 'legacy' and 'advanced' ignore the important technical characteristics of each type of device.

Policy discussions are commonly focused on the node size or the dimensions (measured in nanometres or nm; one-billionth of a metre) of a chip's transistors³⁸. A chip node size refers to the smallest feature size of a transistor on a single chip. However, this notion has not been true for some time. Until 1995, when chips were 0.35 micrometres or 350 nm, nodes described physical gate length in logic chips (ie the actual, measured physical dimension of a transistor's gate electrode, which controls electron flow). Today, node size is more of a marketing label for a generation of technology.

According to the 2022 US CHIPS and Science Act, 'advanced' chips are typically those devices for which the transistor's smallest nodes are approximately under seven to five nm. 'Legacy' chips typically have transistor nodes sizes greater than 28 nm. It should be noted that the notion of an 'advanced' chip is a moving standard. What was considered advanced a decade ago, is considered a legacy chip today. Thus, any sectoral agreement focused on advanced computing must be able to address semiconductor obsolescence to prevent the agreement's product scope from becoming irrelevant. Additionally, the US CHIPS and Science Act device does not define 16nm, 14nm, 12nm and 10nm semiconductor devices as either advanced or legacy and categorically are not defined in either category.

Furthermore, both these labels focus on 'digital logic' devices: devices that are data processing units (ie CPUs, GPUs), though the term digital logic has been expanded to include high bandwidth memory (HBM) chips as well³⁹. Most leading-edge computing systems, including AI systems, use advanced logic chips with an advanced feature size of 5nm/4nm. Even in the most advanced computing systems, however, there are collections of various chip types, including analogue legacy chips. We focus on advanced chips that are crucial in leading-edge computing systems: AI and machine learning, high-performance computing (HPC), advanced graphics and gaming, data centres.

In assessing strategic computing assets, it is essential to define a chip's technical features, the capabilities of the devices it powers and how those devices are used. Chips, whether logic or analogue, advanced or legacy, serve specialised computing functions. Legacy chips, while once state-of-the-art, are now refined and cost-efficient, produced on

Any sectoral agreement focused on advanced computing must be able to address semiconductor obsolescence

38 See for example U.S. Department of Commerce Bureau of Industry and Security, 'Public Report on the Use of Mature-Node Semiconductors', December 2024, <https://www.bis.gov/media/documents/public-report-use-mature-node-semiconductors-december-2024>.

39 Dylan Patel, Myron Xie, Tanj Bennett, Ivan Chiam and Jeff Koch, 'Scaling the Memory Wall: The Rise and Roadmap of HBM', *Semianalysis*, 12 August 2025, <https://newsletter.semianalysis.com/p/scaling-the-memory-wall-the-rise-and-roadmap-of-hbm>.

fully depreciated equipment using optimised processes, which has expanded global participation in their production. By contrast, advanced chips remain scarce and difficult to produce, requiring significant technical expertise, proprietary knowledge and high investment, creating substantial barriers to entry and sustained competitive advantage.

To add some context, Nvidia has invested more than \$58.2 billion in research and development since its inception, indicating the scale of investment needed to achieve advanced computing capabilities (Nvidia, 2025). In relation to advanced chips, competitors focus on harnessing specialised hardware architecture features. For example, unlike CPUs, which process tasks sequentially, GPUs contain thousands of small cores that perform thousands of operations simultaneously.

Policies targeted at semiconductor supply chains, such as through the US CHIPS and Science Act, have focused on advanced logic chips because they power cutting-edge technologies including AI, quantum computing and neuromorphic systems.

However, the semiconductor supply chain is highly global and interdependent, with no single country or company controlling all stages of design, development and manufacturing. A major vulnerability is that most advanced GPUs used for AI training are produced in Taiwan, creating a significant concentration risk in the global supply chain.

3.3 The irresistible appeal of semiconductors

From connected devices to smartphones, artificial intelligence and electric vehicles, today's most common devices can point to an integrated semiconductor device as the source of their communication and functional capabilities. Semiconductor innovation has significant potential for economic prosperity. It is projected that the electric/autonomous vehicle market will be valued at approximately \$1.5 trillion by 2030. Grouped with the renewable energy (market valued at approximately \$4.5 trillion), the data centre AI industry (valued at approximately \$8 trillion by 2030) and the aggregated device AI market (valued at approximately \$2 trillion by 2030), the revenue returns for an economy are massive.

Technological dominance in this area will thus result in huge economic and societal advantages. Furthermore, competitiveness in technology sets the stage for leadership in standardisation and leadership in standardisation provides a first-mover opportunity to drive technological innovation. If a country can be first to develop and commercialise technological innovations, it can have a technological advantage that sets the direction for all global technological utilisation (Palacios, 2026).

Adding to the governmental sense of urgency is the increased link between technology and defence applications, as contemporary commercial innovations are integrated into defence areas. This digital transformation moves defence from traditional domains to a connected battlefield, requiring new approaches to maintain national security: integrating commercial innovations such as AI, drones, cloud computing and satellite tech into military operations to create faster, more adaptable and efficient forces, enhancing strategy, logistics and battlefield awareness (Lang *et al*, 2026).

In an era of heightened geopolitical tension and rapid technological advancement, the United States continues to prioritise defence innovation to maintain its global leadership and national security. Funding proposals in the US Department of Defense

Appropriations Act, 2026, include roughly \$148 billion for research, development, test and evaluation (RTD&E) and continued funding support for the US Defense Innovation Unit (DIU), Accelerate the Procurement and Fielding of Innovative Technologies (APFIT) and the Office of Strategic Capital (OSC)⁴⁰.

With the diffusion of semiconductor design, development and manufacturing capabilities across the globe, agile firms are the main source of technological innovation, influencing policymakers to facilitate R&D (basic research, applied research and experimental development), innovation and strategic investment. Harnessing the innovation lifecycle in the globally competitive semiconductor industry is an opportunity to control strategic assets, especially those dual-use technologies at the intersection of trade, technology and security. As a result, the conceptualisation of a trade agreement that covers trade in both tangible and intangible goods in the advanced semiconductor sector, and which factors in the national and economic security aspects, would be a means to facilitate advanced computing innovation opportunities (ie high-value systems, driven by materials engineering, chip architecture innovations and breakthroughs in advance chip packaging).

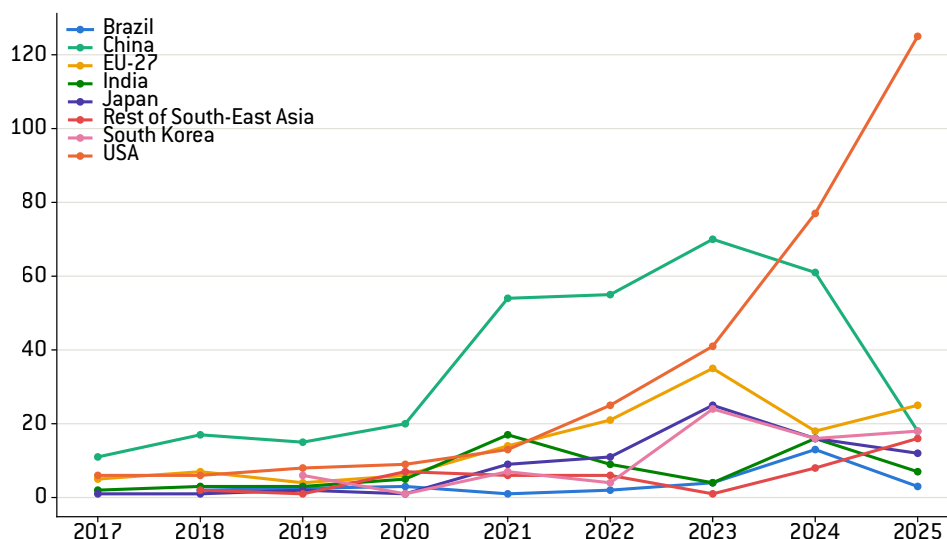
4 Segmenting the world semiconductor market

Data from NIPO⁴¹ confirms that US, China and the EU are the frontrunners by volume on industrial policy measures targeted at semiconductors. Figure 8 shows that US activity is rising. In more modest manner, the EU follows, whereas China has been adopting fewer measures (the overall number of interventions, not their value).

40 See House Committee on Appropriations press release of 18 July 2025, 'House Passes FY26 Defense Bill, Investing in America's Military Superiority', <https://appropriations.house.gov/news/press-releases/house-passes-fy26-defense-bill-investing-americas-military-superiority>.

41 Global Trade Alert (2026). We thank Simon Evenett for allowing us to use his data in the paper.

Figure 8: Industrial policy targeting semiconductors



Source: NIPO.

All the measures counted in Figure 8 may run afoul of the WTO:

- Import duties, to the extent that they exceed the level 'bound' ('capped') during the Uruguay round negotiations (the last concluded round of trade negotiations), violate GATT Article II;
- Import quotas violate GATT Article XI;
- Import licensing schemes must observe the disciplines of the WTO Import Licensing Agreement (ILA), in which non-discrimination features prominently;
- Export quotas violate GATT Article XI;
- Subsidies, as per the current Agreement on Subsidies and Countervailing Measures (SCM) are either illegal (export- and local content subsidies), or countervailable (eg could lead affected parties to impose duties above and beyond the capped level, equivalent to the amount of the subsidy provided). In other words, a WTO member subsidising its domestic production risks that its goods when exported to third countries will in any case be burdened (beyond the agreed customs duty) by an amount equal to the amount of the subsidy paid per unit. Such costs could be higher if domestic subsidies are characterised as export subsidies or as subsidies that include a local-content requirement. In this case, the country could be requested to withdraw the scheme altogether or face retaliation (above and beyond the agreed customs duties) for the amount of damage caused to foreign producers worldwide⁴²;
- The national security ('essential security requirements') exception embedded in GATT Article XXI could in principle justify most (and maybe all) of the above-mentioned violations, but case law has made this impossible:

⁴² Economists have criticised the current WTO regime consistently. Grossman and Sykes (2025) provided an excellent discussion on this issue.

- The absence of any mention of economic security in the body of Article XXI begs the question of the scope of this provision. In DS512 (Russia – Traffic in Transit)⁴³, the panel confirmed the narrow scope of this provision: only measures adopted during war or war-like circumstances could be justified under this provision. This would mean that none of the measures we have listed (see Figure 9) could conceivably come under Article XXI (as constructed in DS512 and all subsequent case law), because of the absence of a war-like scenario;
- Furthermore, it remains unclear whether subsidies can be justified by recourse to this provision at all. Case law has avoided the question of whether an Annex 1A WTO Agreement (like the SCM Agreement) can be justified by a GATT provision.

Despite the various measures employed to promote economic security in relation to semiconductors breaching the WTO Agreement, it is unlikely that trading nations will relinquish their economic security postures, especially now that mistrust has crept in even between the transatlantic partners. Continuation of the current policies will contribute to the further weakening of the multilateral system. But it does not have to be this way. The next section shows how the trend can be reversed will still allowing nations to promote their security objectives.

5 A sectoral agreement on semiconductors

A small but still significant number of trading nations is involved in the production of semiconductors: more for legacy chips, fewer for high-end semiconductors. No country has the 'structural' power to single-handedly impose the regime we outline here, though some might have 'relational' power and be in position to force a certain behaviour on specific targeted nations. The final agreement would have to be an agreement across coalescing forces⁴⁴.

The world trading system has long experience with sectorals⁴⁵. Our proposal works whether or not the membership opts for a critical mass MFN agreement, or for variable geometry, eg an agreement that will bind only signatories, and to which accession will be in function of meeting the requirements met by the original signatories. In our view, a

43 See WTO, 'Russia — Measures Concerning Traffic in Transit', https://www.wto.org/english/tratop_e/dispu_e/cases_e/ds512_e.htm.

44 Strange (1988) distinguished between relational power (the power of A to influence B), and structural power (the power of either A or B to determine the rules and constraints of the game). See also Bachrach and Baratz (1962).

45 The WTO Secretariat issued in 2005 a paper entitled 'Sector Specific Discussions and Negotiations on Goods in the GATT and the WTO', which discussed the history of sectoral agreements that have been concluded since the advent of the GATT, WTO Doc. TN/MA/A/S/13 of 24 January 2005.

critical mass MFN would be preferable to another FTA or sectoral agreement outside the WTO.

5.1 The mechanics

The technical modalities of introducing these agreements into the WTO are as follows.

5.1.1 The process for implementing the suggested sectoral

Recall that by virtue of GATT Article II.7: *"The Schedules annexed to this Agreement are hereby made an integral part of Part I of this Agreement."*

A formal process has been put into place to guide changes in schedules⁴⁶. The 1980 'Decision on Procedures for Modification and Rectification of Schedules of Tariff Concessions'⁴⁷ is the legal instrument that allows for unilateral changes to a schedule (which is what we propose for semiconductors), but subjects them to a multilateral process. Changes can comprise either 'rectifications' or 'amendments'. A rectification does not affect the substance of the negotiated commitment, while an amendment does.

No matter how the WTO member concerned qualifies the change, it must submit it to the WTO Secretariat's Market Access Division. The latter will circulate it to the membership. If no objection arises within three months, it will include the change in the member's schedule (certification). If an objection is raised, it will not. Whatever the case, a dispute can subsequently be submitted, even if there were no objections to the inclusion of a new concession. Certification does not confer legality⁴⁸. It is a process to verify the accuracy of commitments entered into. The legality of commitments that have been certified can be challenged, as can the legality of commitments that have been challenged and consequently have not been certified⁴⁹. WTO members affected by the inclusion of dubious 'entries' in a national schedule can protect their interests even if they did not react when the certification process was underway, and even if it was completed without them raising an objection.

5.1.2 Scope of the sectoral

The trading nations could opt for one of two possible scenarios. The expansive scenario would be to write an agreement for trade in all semiconductors. The narrower design would limit product coverage to advanced semiconductors only. High-end semiconductor capabilities are defined by cutting-edge performance features, advanced manufacturing processes and the use of specialised materials (via advances in material science) to enable high compute-demanding applications such as AI, autonomous systems and high-performance computing. The membership affected would depend on which of the two alternatives is privileged, as only a handful of nations produce the advanced goods. Consider the following key segments:

⁴⁶ What now follows borrows heavily from Hoekman and Mavroidis (2017).

⁴⁷ GATT Doc. BISD 27S/25-26.

⁴⁸ The Appellate Body in EC-Bananas III ruled as much; see also Santana and Jackson (2012).

⁴⁹ Hoekman and Mavroidis (2017) discussed the relevant case law. This approach was followed for the implementation of the ITA and the Joint Initiative on Services Domestic Regulation.

Table 1: Leading suppliers in advanced semiconductor segments

Semiconductor Sector	Leading Suppliers
Foundry fabrication	Taiwan (TSMC), S. Korea (Samsung), US (Intel)
Equipment & materials	Lithography: EU-Netherlands (ASML), Japan (Nikon, Canon) Process Equipment: US (Applied Materials, LAM, KLA), Japan (TEL), Netherlands (ASM) Materials: Japan (Shin-Etsu Chemical, JSR)
Fabless design	USA (Nvidia, AMD, Broadcom), Taiwan (Mediatek)
Memory DRAM (high bandwidth memory)	Republic of Korea (SK Hynix, Samsung), US (Micron)

Source: Bruegel.

Restricting the membership (at the initial stage) would facilitate the negotiation of the arrangement. However, ‘wider’ product coverage could be seen as more legitimate, especially because more trading nations would be asked to condone the new ‘economic security’ clause that, in our view, should be an essential element of the agreement.

Our preference is for an agreement limited to advanced semiconductors. Legacy semiconductors are available practically everywhere and the design, development and manufacturing capabilities for these devices are spread across a wide swath of countries, taking advantage of mature fabrication technologies. There is a huge market for re-use/re-furbished manufacturing equipment, which has facilitated entry into this market with less R&D cost compared to advanced chips. Legacy chip firms focus on maintaining stable, high-volume production for essential, low-cost applications using older, well-established manufacturing processes with cheaper, widely available equipment.

Plus, to our knowledge, trade in legacy chips is not restricted at all, especially for general-purpose commercial civilian devices. Furthermore, importing countries do not burden them with tariffs, because of the Information Technology Agreement (ITA) and ITA 2⁵⁰.

However, there could be some consideration made for legacy devices that feature new innovative capabilities, incorporating advanced materials such as silicon carbide, or gallium nitride. But like devices used in advanced computing, opening the potential agreement to a specific set of legacy chips should be focused on specific compute uses: optoelectronics, power electronics and high-frequency electronics. However, as mentioned above, trade in these devices is not generally subject to disruptive trade restrictions and is not affected by national and economic security concerns.

Conversely, advanced semiconductors present with a host of issues that have caused much acrimony, especially in relation to national security. Dual-use chips link to advanced computing and the inherent economic and national defence significance of these capabilities to countries globally. Additionally, the massive R&D expenditure on advanced chips is driven by the relentless challenge of Moore’s Law (or what’s left of it: *“chipmakers are relying more and more on exotic materials, advanced packaging technologies and more complex 3D transistor designs to deliver improved chip performance”*⁵¹) and the

50 See WTO, ‘Information Technology Agreement’, https://www.wto.org/english/tratop_e/dtt_e/dtt-ita_e.htm.

51 See ASML, ‘Moore’s Law’, undated, <https://www.asml.com/en/technology/all-about-microchips/moores-law>.

intense global competition to innovate at nanoscale. A realpolitik-inspired negotiation in this narrow context could prove beneficial for a more general updating of GATT Article XXI.

Our recommendation of an agreement on advanced semiconductors will of course have a bearing on the identity of the negotiators. The following economies have a legacy semiconductor supply-chain connection: the US, EU, Japan, Korea, Taiwan, Canada, China, India, Israel, Malaysia, New Zealand, Norway, Thailand, Indonesia, Singapore, Switzerland, Vietnam and the Philippines. Only a few (Canada, the US, EU, India, Israel, Japan, Korea, Taiwan) possess high-end chip capability.

China does not currently possess the indigenous technology, hardware and equipment to fully fabricate an advanced logic chip (less than 7-5nm). Chinese foundries leverage western equipment (ie lithography from the Netherlands, etch and deposition from Japan) for 7nm production. While China is rapidly advancing its DRAM industry, with Changxin Memory Technologies (CXMT) producing 16nm DDR5, narrowing the technology gap with global leaders, it remains heavily reliant on western semiconductor manufacturing equipment.

5.1.3 MFN trade

Signatories accept that the end product of their negotiations should be subject to MFN. This should be an issue only with respect to the producers that refuse to accept commitments. The problem could be circumvented by introducing coverage of a high percentage of world production as a pre-condition for the entry into force of the arrangement ('critical mass'). Signatories should agree to continue to import duty-free all covered products, except for those listed as sensitive on national-security (and economic-security) grounds. Each signatory should be allowed to insert a fixed number of products into this list.

This raises a legal question of interest to ITA signatories. The ITA already covers some semiconductors, though not the most recent that have been developed since the conclusion of the ITA. For this latter category, no issues arise at all. But legal issues could arise for those semiconductors already covered by ITA and ITA 2 with respect to their tariff treatment.

The non-tariff elements in ITA I and II are limited in scope and cover the guidelines on electromagnetic compatibility and interference, not the issues we discuss here. But under our proposal, a signatory might include a product already covered by the ITA (I or II) in its list of security exceptions. The effects on signatories of both the ITA and our proposed agreement, and signatories of only the ITA, should be distinguished. The latter category will have agreed to the new rules and by virtue of *lex posterior* (subsequent agreement) will have no claim for compensation as they have voluntarily agreed to that. The former though could have a legitimate claim for compensation as they might be hurt by subsidies (for which we propose a safe haven), or measures restricting trade.

As far as the subsidies component of our proposal is concerned, countries will have to show damage on the basis of demonstration that high-end and legacy semiconductors are substitutes with respect to specific functions. This could very well be the case. Signatories of our proposed agreement, when exporting subsidised high-end semiconductors could

face countervailing duties (CVDs). This is unlikely, as importers would have an interest in importing similar products, but possible. When shunning legacy semiconductors from their markets (assuming always that they compete with legacy semiconductors originating in ITA signatories), signatories might be called to pay compensation. But as time passes, and industry switches progressively to high-end products, this will become less likely. Nevertheless, to avoid this eventuality, we recommend a bargaining solution with the ITA signatories that do not join the proposed agreement.

When it comes to justifying exports, signatories of the proposed agreement, without a re-orientation of current case law, will find it hard to justify their measures under GATT Article XXI. Consequently, export-restricting measures could be found inconsistent with WTO rules.

Once again, unless if a bargaining solution has been reached with the signatories of the ITA, those adopting similar measures will have to accept the imposition of countermeasures. Note, nevertheless, that the quantification of damage will be very hard for the plaintiffs, as they will have to quantify the damage resulting from the impossibility to import high-end semiconductors.

With this in mind, we turn, for our purposes, to the understanding of the term 'economic security' (see also section 2).

Economic security has been defined in some national statutes. For example, a US domestic statute defines it as meaning: *"...the condition of having secure and resilient domestic production capacity, combined with reliable access to the global resources necessary to maintain an acceptable standard of living and to protect core national values"*⁵².

In similar vein, the Council on Foreign Relations (2025) defined economic security as:

"...anything related to the economy that affects national security. That includes, for example, research and development (R&D), the defense industrial base, and essentially any government policy that significantly shapes the contribution of the economy to national security. Since the US economy underpins diplomatic and military power, an overly broad approach to economic security can quickly become impractical, as most economic activities touch on national security in some way".

This is an open-ended definition, but a very plausible one. Economic security is a context-dependent term that is elastic by construction. The US definition quoted above could of course provide useful inspiration for a definition that could find its way within the WTO. However, it would be wise if the WTO membership avoided defining the term 'economic security'. We have no reliable metric of the term 'economic security' in this context. Some nations are more risk averse than others; some have larger and some smaller bargaining power. Some have reliable allies, and some do not. To come up with one common definition would be difficult. Thus, the term should rather become a negotiated instrument than a matter for definition.

Furthermore, as noted in Mavroidis (2025), parties supplying evidence about

52 6 USC§ 474 (c)(2).

security might legitimately be disincentivised from doing so, especially because of the WTO's inadequate judicial framework when it comes to guaranteeing the security of confidential information. And of course, the risks of 'incomplete' definitions should not be underestimated; the 'incompleteness' of Article XXI led the adjudicators in DS512 (the 'leading' case in the context; see section 4) to narrow the applicability of Article XXI to war and war-like situations only. This formula has been replicated in all subsequent case law.

It would be more sensible to negotiate a list of dual-use goods, agreed by all WTO members, for which the uninhibited trade would present an unacceptable risk⁵³. The list would include products on which restrictions (import, export) could be legitimately placed⁵⁴. The list should of course be periodically updated to reflect modern realities⁵⁵. The goods to be included should be those that each signatory to our proposed agreement considers necessary in order to achieve its national and/or economic-security objectives.

The negotiation should aim to minimise the potential for active industrial policy under the guise of promoting economic security. To this effect, we suggest a two-step procedure:

1. First, ensure that products included in the list are genuine dual-use (eg industrial, military) goods. Some semiconductors, such as legacy node logic chips, older memory chips (DRAM using a node technology above 18 nanometres), low-density storage (eg NAND flash memory with fewer than 128 layers), consumer electronic components (eg sensors or standard computing peripherals), automotive/industrial chips (standard analogue chips, micro-controllers) and even basic components (diodes, transistors, passive components) are not dual-use goods. They need not be part of the agreement.
2. Second, perform an additional sorting across the remaining semiconductors depending on the availability of the products. It would make little to include in the agreement products that enjoy wide availability. What constitutes 'wide' or 'sufficient' availability is something the negotiators could usefully agree.

A less ambitious (but maybe more acceptable approach) would be for the membership to prepare an open-ended definition of economic security complemented by an indicative list of products for which restrictions would be appropriate to serve economic security. The indicative list would help those entrusted with the enforcement of the agreement to avoid type II errors (false negatives), whereas the definition would help prevent circumvention: inspired by the indicative list, the members, in case of dispute, could add to the list.

53 See the relevant discussion in Mavroidis (2025).

54 Of course, to the extent that a WTO member has not bound them, it remains free to impose export duties, provided that it observes the MFN principle. Lester and Zhu (2019) were the first to table a proposal along these lines – that is, to make security a negotiable instrument.

55 National governments have been working with national lists for some time, so there is already a basis for an international negotiation in this context. See for example the 16 critical infrastructure sectors identified by US National Security Memorandum 22, 'Critical Infrastructure Security and Resilience', 30 April 2024, <https://www.presidency.ucsb.edu/documents/national-security-memorandum-critical-infrastructure-security-and-resilience>.

This innovation should qualify as WTO-inconsistent. Why so? Simply because the signatories of our proposed agreement will be withdrawing advantages granted under the ITA, under which they promised the WTO membership elimination of tariffs for semiconductors and semiconductor-manufacturing equipment on an MFN-basis. As per GATT Article XXVIII, bound duties cannot be increased unilaterally. Compensation needs to be paid to those negatively affected. But those affected will be part of the negotiation anyway.

Finally, negotiators could be inspired by the EU's practice of determining 'adequacy' in relation to data protection⁵⁶. The European Commission can determine whether a non-EU country offers an adequate level of data protection (equivalent to that of the EU), providing a legal basis for transfers of data. Similarly, signatories to the agreement could exonerate some participants from trade restrictive measures, provided they do so based on transparent and verifiable criteria. MFN would then apply against this background, that is, if beneficiaries can adequately take care of the concerns of exporting nations. In other words, 'trusted' partners will not be subjected to export restrictions.

Each signatory will thus have the right to include a limited number of listed products (HS classifications). Restrictive measures can be taken by observing a transparency obligation to this effect, that is, by explaining which measure has been adopted and with respect to which listed product. The list of HS classifications to be included could be inspired by the list of critical technologies developed by the US Office of Science and Technology Policy and other similar entities. The term 'critical technologies' originates in the so-called Bucy report from 1976 (Defense Science Board Task Force, 1976). This understood the term to roughly cover companies that produced sophisticated technical data, and recommended that restrictive measures to promote security should be confined to products with these characteristics, though applying the definition proved difficult in practice. The periodic updating of the list we propose will ensure that it does not become outdated. To avoid confusion, we propose that each member issues one list and it is this list that it will be updating through periodic re-negotiations.

Every other restrictive measure will have to be justified. The listed products will be restricted on grounds of economic security, whereas the national security clause could be invoked for all products covered by the proposed agreement that have not been included in the list that each signatory to our proposed agreement has issued; for these, restrictions can be lawfully imposed. The national security clause should be non-justiciable (Mavroidis, 2025). Similar issues are better handled by committees behind closed doors (in light of the confidentiality of issues involved, and probably even reputation costs in case of abuse). Furthermore, besides disagreements between parties over the content of the term 'national security', the inherent incompleteness of the term implies a high risk of judicial error, as per Wolfers (1962):

*"When **political formulas** such as 'national interest' or 'national security' gain popularity they need to be scrutinized with particular care. They may not mean*

56 See European Commission, 'Adequacy decisions', https://commission.europa.eu/law/law-topic/data-protection/international-dimension-data-protection/adequacy-decisions_en.

the same thing to different people. They may not have any precise meaning at all. Thus, while appearing to offer guidance and a basis for broad consensus, they may be permitting everyone to label whatever policy he favors with an attractive and possibly deceptive name” [emphasis added]⁵⁷.

Scrutiny yes, judicialisation no, is the essence of our recommendation. The common interest of trading nations will be better served if they rekindle the management of their disputes (as during the GATT years) instead of rushing to adjudicate ultra-sensitive issues such as national security. Of course, as things stand, GATT Article XXI is justiciable.

5.1.4 Subsidies

With respect to subsidies, our suggested agreement should state that:

- It takes precedence over the generic SCM Agreement, as far as semiconductor subsidies are concerned;
- Unlike the SCM Agreement, it provides for a ‘safe haven’ for some subsidies. There are significant knowledge spillovers and, without the possibility for subsidies, a coordination failure (‘hold-up’ problem) might arise, as rational investors might be unwilling to spend money that will profit free riders. A cap reflecting a realistic amount dedicated to R&D subsidies should thus be introduced to guarantee a level playing field across all participants;
- Similarly, a provision should be made for production that observes environmental norms (either unilaterally designed, or multilaterally agreed). The semiconductor industry is a concern for **environmental protection related to** its intensive use of energy, water and hazardous chemicals, from raw-material extraction to manufacturing and end-of-life disposal. To the extent that signatories are willing to subsidise the cost associated with pollution, they should be allowed to do so;
- Include a process to scale the product scope, which is focused on keeping pace with chip innovation to ensure that the advanced semiconductors included in the agreement, plus the manufacturing equipment used to fabricate them, are genuinely those considered ‘advanced’ by industry. As today’s high-end goods become ‘legacy’ over time, they will be excluded from national lists allowing for restrictive measures.

5.1.5 Export restrictions

Negotiators should include an export licensing agreement (ELA) that will replicate the existing import licensing agreement (ILA) disciplines. Export restrictions are increasingly seen either in finished goods (semiconductors) or in inputs to the finished product (rare earths). An ELA will help streamline restrictions, just as the ILA has so far⁵⁸. Our proposed ELA should apply to all products not included in the ‘security’ list that we propose. As is

⁵⁷ Wolfers (1962) went on to quip that national security is of course the cardinal preference of nations and associated policies subordinate other interests to that of the nation itself. He concluded that, beyond that, the term has very little meaning.

⁵⁸ Mavroidis (2025) discusses this point in detail.

the case with the ILA, conformity with the ELA will of course be justiciable. Consequently, signatories wishing to impose restrictions above and beyond the security list, will have to comply with the disciplines embedded in the proposed ELA.

5.1.6 Enforcement

Disputes about the practice of national security (including economic security) can arise, as nations have differing views on this issue. It would make better sense to entrust such discussions to a WTO committee in which delegates would discuss and, if needed, agree to disagree. They should learn to manage and not necessarily resolve their disputes. Enforcement of R&D spending caps and/or the improper inclusion of items in lists of 'sensitive' goods should be points for discussion. Conversely, there should be no room for adjudication that challenges trade-restricting measures on goods that have been properly included in the 'sensitive' list. The review of the agreement will be monitored by the signatories of the agreement only.

Our proposed agreement could provide a framework within the WTO for trade in sensitive, dual-use goods. The intention is to enable the uninhibited pursuance of plausible objectives while providing relative certainty to the world trading community about the rules.

Table 2: A schematic representation of our proposal

Coverage	Participants	MFN	Security exception	Export restrictions	Subsidies
High-end semiconductors	All producers of high-end semiconductors	All high-end goods covered	Negotiated based on our two-step procedure	Disciplines replicate the WTO ILA	R&D expenditure capped

Source: Bruegel.

5.2 What our proposal can achieve

In previous club agreements, the private sector has played a major role in the negotiation and the implementation of the final deal. Engagement with industry can take the form of indirect consultations with industry groups and associations, or directly with company representatives. Such engagement helped both the original ITA I and the expanded ITA II to reach successful conclusions. But it is one thing to agree a text. Implementation must also be adequate.

Industry involvement can lead to pro-liberalisation or pro-protectionism outcomes. The Multi-Fiber Agreement⁵⁹, for example, was clear a market-segmenting agreement, whereas the ITA is a liberalisation device. Our suggested agreement is clearly a liberalization device, as it aims to bring order to the unilateral measures that hamper trade in semiconductors.

Ideally of course, the agreement should be 'wide', with disciplines on inputs into

59 GATT Doc. COM.TEX/W/102 of 15 July 1981, available at https://www.wto.org/gatt_docs/English/SULPDF/90630047.pdf.

semiconductors (rare earths) also part of the negotiation. This however would require the simultaneous presence of the US and China in negotiations, which is why it is not a make-or-break issue in our proposal. If the US and China did agree to that, their participation would substantially increase the legitimacy of the end agreement.

Product coverage could be delimited by introducing the HS classifications of the products concerned, or by describing some of the products (as it might prove necessary to extend the coverage to technology that goes beyond the strict definition of a 'semiconductor'), or both, eg include HS classifications whenever possible and act based on to descriptions of items when necessary.

On the proposed subsidy discipline, two potential criticisms arise. First, how can a part of the WTO membership impose a relaxation of the subsidy discipline on the WTO membership? Second, why is a relaxation necessary? We propose a relaxation of the subsidy regime to facilitate the introduction of a safe haven for R&D subsidies (to encourage innovation, address knowledge spillovers, etc). We are actually proposing a re-introduction of this category of subsidies, as the original Article 8 of the SCM made room for them. But Article 8 of the SCM expired after five years (SCM Article 31) and was not renewed. Why is it necessary to re-introduce a provision along these lines with more realistic sums involved this time? After all, one might legitimately argue that similar subsidies are only 'actionable'⁶⁰. Consequently, those who object to subsidies can at best request a retreat from subsidisation. But of course, this risk is close to zero, as all producers of semiconductors will have signed our proposed agreement. And why would consumers (who, if at all, will profit from the lower prices resulting from the subsidisation) challenge the legality of relaxing the current discipline?

The real risk originates in the 'completion' of the contract through case law. The agency design for adjudicating disputes in the WTO is insufficient. There are numerous examples of unsatisfactory results (Mavroidis, 2022). If the door to 'permissible' subsidies is closed, WTO panels might blur the distinction between 'actionable' and 'prohibited'. The evidentiary standard for the latter is very low, and opportunistic members might indulge in litigation for purely selfish reasons (eg to fend off legitimate claims from subsidisers of semiconductors about other issues). Errors will not be confined to the jurisprudence regarding semiconductors, but will expand to the understanding of the SCM Agreement.

Critics might also find our proposal too deferential to national qualifications of security. But the counterfactual is the current situation, in which panels routinely interpret GATT Article XXI in roughly the same way, and culprits refuse to implement the final reports. Furthermore, the current interpretation of this provision makes no room for promotion of economic security, a concept distinct from essential security interests (the term appearing in the GATT statute). Our proposal, if taken seriously, will lead to a negotiation of items for which restrictions are permissible thus providing some certainty on trade in semiconductors.

Our solution is inspired by the negotiation of the original GATT. Faced with the question of how to define 'protection' in order to tame it, the negotiators, instead of defining it, adopted a less-ambitious but functional attitude: they reduced 'protection' to one

60 We owe this argument to Jeromin Zettelmeyer, who pushed us to think this issue through.

negotiable instrument, namely, tariffs. They outlawed quantitative measures at the border and obliged members to apply their behind-the-border policies in a non-discriminatory manner. Consequently, trading nations wishing to protect domestic producers could only do so through tariffs.

Similarly, instead of attempting to define 'economic security', we suggest that WTO members should reduce it to a negotiable instrument. Each of them would draw up a list of goods that could be subjected to trade-restrictive measures. The remaining product markets could be protected only through the existing contingent protection instruments.

And anyway, it is not easy to imagine how one could go beyond our framework without pushback. Nations are unwilling to compromise economic security. The reigning uncertainty has divided even staunch allies, such as the transatlantic partners. Those who see parallels in history compare modern times with the situation before the First World War (Westad, 2026). Our proposed agreement is an effort to re-establish channels of communication across nations globally. There are ways and means to advance the concerns of trading nations without moving outside the disciplines embedded in the multilateral edifice.

In this respect, we would like to underscore the potential positive external effects of our recommendation: if it works, trading nations might wish to re-engage with the multilateral machinery once again. This would be the biggest gain for the world community.

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Bruegel, Rue de la Charité 33, B-1210 Brussels
(+32) 2 227 4210
info@bruegel.org
www.bruegel.org